MMDU MULLANA OFFICIAL SYLLABUS • BMAT-001 • UNIT 3

Unit 3: Unit III: Multivariable Calculus (Differentiation) (10 Hours)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 35 mins

Complete Unit III coverage per MMDU BMAT-001 syllabus: Limit, continuity and partial derivatives, Homogeneous Function, Euler's theorem, total derivative; Maxima and minima for the function of two variables; Method of Lagrange multipliers. Scalar and Vector functions, Gradient, Directional derivatives, divergence and curl.

1. Limits, Continuity & Partial Derivatives

1. Multivariable Functions: Limits, Continuity & Partial Derivatives

A. Limit of a Function of Two Variables:

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous ϵ-δ Definition: We write $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

- **Path Independence Principle:** For a limit to exist in R^2, the function must approach the **exact same value** L **along EVERY conceivable path** approaching (a, b).
- **Two-Path Non-Existence Test:** If $f(x,y)$ yields different values along paths $y = mx$ or $y = mx^2$ (i.e., limit depends on m), the limit **strictly DOES NOT EXIST**!

B. Continuity of $f(x,y)$:

A function $f(x,y)$ is continuous at (a,b) if:

1. $f(a,b)$ is defined.
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.
3. $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

2. Partial Derivatives & Higher-Order Derivatives

$$\frac{\partial f}{\partial x} = f_x(x,y) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}, \quad \frac{\partial f}{\partial y} = f_y(x,y) = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

✅ **FUNDAMENTAL LAW / RULE:**

Clairaut's Symmetry Theorem: If the mixed partial derivatives f_{xy} and f_{yx} are both **continuous** on an open region containing (a, b), then:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \iff f_{yx} = f_{xy}$$

University Question (4 Marks):

Evaluate whether $\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4+y^2}$ exists.

Step 1 🚀: Approach along straight line paths $y = mx$

$$\lim_{x\to 0} \frac{x^2(mx)}{x^4 + m^2x^2} = \lim_{x\to 0} \frac{mx}{x^2 + m^2} = 0$$

Step 2 ⚡: Approach along parabolic path $y = mx^2$

$$\lim_{x\to 0} \frac{x^2(mx^2)}{x^4 + m^2x^4} = \frac{m}{1+m^2}$$

Since the limit depends on m, the limit is **not unique** and **does NOT exist**.

🎯 By Path Test ($y = mx^2$), the limit **does NOT exist**.

University Question (6 Marks - MMDU Question Paper):

If $u = \ln(x^2 + y^2)$, prove that u satisfies 2D Laplace's Equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Step 1 🚀: First Partial Derivatives

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2+y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2+y^2}$$

Step 2 ⚡: Second Partial Derivatives via Quotient Rule

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(x^2+y^2)-2x(2x)}{(x^2+y^2)^2} = \frac{2y^2-2x^2}{(x^2+y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2x^2-2y^2}{(x^2+y^2)^2}$$

Step 3 🎯: Sum the Second Derivatives

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2y^2-2x^2+2x^2-2y^2}{(x^2+y^2)^2} = 0$$

🎯 Proved: $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (u is Harmonic).

2. Euler's Theorem on Homogeneous Functions

1. Homogeneous Functions & Euler's Theorem: Rigorous Analysis

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A multivariable function $f(x,y)$ is **homogeneous of degree** n if for any scalar $t > 0$:

$$\mathbf{f(tx, ty) = t^n f(x, y) \iff f(x, y) = x^n \phi\left(\frac{y}{x}\right)}$$

2. Euler's Theorem on Homogeneous Functions (First & Second Order)

✅ **FUNDAMENTAL LAW / RULE:**
First-Order Euler's Theorem: If $u = f(x,y)$ is a homogeneous function of x and y of degree n with continuous partial derivatives, then:

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=n\,u$$

✅ **FUNDAMENTAL LAW / RULE:**
Second-Order Euler's Formula:

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=n(n-1)u$$

🚨 **EXAMINER TRAP / PITFALL:**
Examiner's Composite Function Deduction: If $z=f(u)$ is homogeneous of degree n in (x,y), then:

1. $xu_x+yu_y=n\frac{f(u)}{f'(u)}=g(u)$

2. $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=g(u)\,[g'(u)-1]$

University Question (6 Marks - MMDU Question Paper):

If $u=\sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$, prove that $xu_x+yu_y+zu_z=-3\tan u$.

Step 1 🚀: Convert to Homogeneous Function $v=\sin u$

$$v=\sin u=\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$$

Degree of numerator is 1; degree of denominator is $\sqrt{t^8}=t^4$. Thus v is homogeneous of degree $n=1-4=-3$.

Step 2 ⚡: Apply 3-Variable Euler's Theorem to v

$$x\frac{\partial v}{\partial x}+y\frac{\partial v}{\partial y}+z\frac{\partial v}{\partial z}=-3v$$

Using $\frac{\partial v}{\partial x} = \cos u \frac{\partial u}{\partial x}$:

$$\cos u\,(xu_x + yu_y + zu_z) = -3\sin u \implies xu_x + yu_y + zu_z = -3\tan u$$

🎯 Result Proved: $\mathbf{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = -3\tan u}$.

University Question (6 Marks - MMDU Previous Paper):
If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$.

Step 1 🚀: Define $z = \tan u$ (Degree $n = 3 - 1 = 2$)

$$xz_x + yz_y = 2z \implies x\sec^2 u\,u_x + y\sec^2 u\,u_y = 2\tan u \implies xu_x + yu_y = \frac{2\tan u}{\sec^2 u} = \sin 2u = g(u)$$

Step 2 ⚡: Second-Order Euler Identity

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1] = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$$

🎯 Second-Order Euler Identity Proved: $\mathbf{x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(2\cos 2u - 1)}$.

3. Total Derivative, Maxima-Minima & Lagrange Multipliers

1. Total Differential, Chain Rule & Extrema of Multivariable Functions

A. The Total Differential (du) & Total Derivative ($\frac{du}{dt}$):

- **Total Differential:** $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$
- **Total Derivative (Single Parameter t):**

$$\frac{\mathbf{du}}{\mathbf{dt}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\mathbf{dx}}{\mathbf{dt}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\mathbf{dy}}{\mathbf{dt}}$$

- **Composite Chain Rule ($x = x(r,\theta),\ y = y(r,\theta)$):**

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}, \quad \frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

---

2. Unconstrained Maxima & Minima of Two Variables ($rt - s^2$ Test)

To locate the local maxima, local minima, and saddle points of $z = f(x, y)$:

1. **Stationary Points:** Solve $f_x = 0$ and $f_y = 0$.

2. **Second-Order Partials at (a, b):** $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$ and Discriminant $\Delta = rt - s^2$:

Condition on $\Delta = rt - s^2$	Condition on $r = f_{xx}$	Nature of Critical Point (a, b)
$rt - s^2 > 0$	$r < 0$	**Local Maximum**
$rt - s^2 > 0$	$r > 0$	**Local Minimum**
$rt - s^2 < 0$	Any sign	**Saddle Point**
$rt - s^2 = 0$	Indeterminate	**Test Fails** (Further investigation required)

---

3. Constrained Optimization: Method of Lagrange Multipliers

To optimize $f(x, y, z)$ subject to constraint $\phi(x, y, z) = 0$:

1. Form Auxiliary Lagrangian: $F(x, y, z, \lambda) = f(x, y, z) + \lambda\, \phi(x, y, z)$.

2. Solve the stationary system:

$$f_x + \lambda \phi_x = 0, \quad f_y + \lambda \phi_y = 0, \quad f_z + \lambda \phi_z = 0, \quad \phi(x, y, z) = 0$$

University Question (4 Marks):
Find the local maximum and minimum values of $f(x, y) = x^3 + y^3 - 3xy$.

Step 1 🚀: Find Critical (Stationary) Points
$f_x = 3x^2 - 3y = 0 \implies y = x^2$ and $f_y = 3y^2 - 3x = 0 \implies 3x(x^3 - 1) = 0$.
Critical points: $(0, 0)$ and $(1, 1)$.

Step 2 ⚡ : Compute Second Partials r, s, t

$r = f_{xx} = 6x, \quad s = f_{xy} = -3, \quad t = f_{yy} = 6y \implies \Delta = rt - s^2 = 36xy - 9.$

Step 3 🎯 : Classify Points

1. **At** $(0,0)$**:** $\Delta = 0 - 9 = -9 < 0 \implies (0,0)$ is a **Saddle Point**.

2. **At** $(1,1)$**:** $\Delta = 36 - 9 = 27 > 0$ and $r = 6 > 0 \implies (1,1)$ is a **Local Minimum** with $f_{\min}(1,1) = 1 + 1 - 3 = -1$.

🎯 Saddle point: $(0,0)$ | Local Minimum at $(1,1)$ with $\mathbf{f_{min} = -1}$.

University Question (6 Marks - MMDU Question Paper):
Find the maximum volume of a rectangular box open at the top, having a given surface area $S = 108\text{cm}^2$, using Lagrange's method of multipliers.

Step 1 🚀 : Objective and Constraint Functions
Maximize $V = xyz$ subject to $\phi(x,y,z) = xy + 2yz + 2zx - 108 = 0$.
Lagrangian: $F(x,y,z,\lambda) = xyz + \lambda(xy + 2yz + 2zx - 108)$.

Step 2 ⚡ : Set First Partials to Zero

$$yz + \lambda(y + 2z) = 0, \quad xz + \lambda(x + 2z) = 0, \quad xy + \lambda(2x + 2y) = 0$$

Equating λ gives $x = y = 2z \implies z = \frac{x}{2}$.

Step 3 🎯 : Substitute into Constraint $S = 108$

$$x^2 + 2x(x/2) + 2x(x/2) = 3x^2 = 108 \implies x = 6,\ y = 6,\ z = 3 \implies V_{\max} = 6 \times 6 \times 3 = 108\text{cm}^3$$

🎯 Dimensions: $6\text{cm} \times 6\text{cm} \times 3\text{cm}$ | Maximum Volume: $\mathbf{V_{max} = 108\text{cm}^3}$.

4. Scalar & Vector Functions, Gradient, Divergence & Curl

1. Vector Differential Calculus: Del, Gradient, Divergence & Curl

✅ **FUNDAMENTAL LAW / RULE:**
The Vector Differential Operator Del (∇):

$$\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial \mathbf{x}} + \hat{\mathbf{j}}\frac{\partial}{\partial \mathbf{y}} + \hat{\mathbf{k}}\frac{\partial}{\partial \mathbf{z}}$$

A. Gradient of a Scalar Field ($\nabla\phi$) & Directional Derivative:

- **Gradient:** $\nabla\phi = \hat{\mathbf{i}}\frac{\partial\phi}{\partial \mathbf{x}} + \hat{\mathbf{j}}\frac{\partial\phi}{\partial \mathbf{y}} + \hat{\mathbf{k}}\frac{\partial\phi}{\partial \mathbf{z}}$ (normal to level surface $\phi(x, y, z) = c$).
- **Unit Normal Vector:** $\hat{\mathbf{n}} = \frac{\nabla\phi}{|\nabla\phi|}$.
- **Directional Derivative along $\vec{a}$:** D.D. $= \nabla\phi \cdot \frac{\vec{\mathbf{a}}}{|\vec{\mathbf{a}}|}$ (maximum magnitude = $|\nabla\phi|$ in direction of $\nabla\phi$).

B. Divergence ($\nabla \cdot \vec{F}$) & Curl ($\nabla \times \vec{F}$):

- **Divergence (Scalar):** $\text{div}\,\vec{\mathbf{F}} = \nabla \cdot \vec{\mathbf{F}} = \frac{\partial \mathbf{F_1}}{\partial \mathbf{x}} + \frac{\partial \mathbf{F_2}}{\partial \mathbf{y}} + \frac{\partial \mathbf{F_3}}{\partial \mathbf{z}}$.

- **Solenoidal Field:** $\nabla \cdot \vec{\mathbf{F}} = \mathbf{0}$.

- **Curl (Vector):** $\text{curl}\,\vec{\mathbf{F}} = \nabla \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$.

- **Irrotational (Conservative) Field:** $\nabla \times \vec{\mathbf{F}} = \vec{\mathbf{0}} \iff \vec{\mathbf{F}} = \nabla\phi$.

University Question (6 Marks - MMDU Question Paper):
Find the directional derivative of $\phi = x^2yz + 4xz^2$ at point $P(1, -2, -1)$ in the direction of vector $\vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Step 1 🚀: Calculate $\nabla\phi$ at $P(1, -2, -1)$

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\Big|_{(1,-2,-1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Step 2 ⚡: Project onto Unit Vector $\hat{a}$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{4 + 1 + 4}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \implies \text{D.D.} = \nabla\phi \cdot \hat{a} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

🎯 Directional Derivative: $\frac{37}{3} \approx \mathbf{12.33}$.

University Question (6 Marks - MMDU Question Paper):
Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is irrotational, and find its scalar potential ϕ.

Step 1 🚀: Evaluate $\text{curl}\,\vec{F} = \nabla \times \vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix} = (0)\hat{i} - (3z^2 - 3z^2)\hat{j} + (2y \cos x - 2y \cos x)\hat{k} = \vec{0}$$

Step 2 ⚡: Integrate Exact Differential $d\phi = \vec{F} \cdot d\vec{r}$

$$d\phi = d(y^2 \sin x) + d(xz^3) - d(4y) + d(2z) \implies \phi = y^2 \sin x + xz^3 - 4y + 2z + C$$

🎯 $\nabla \times \vec{F} = \vec{0}$ (Irrotational) | Scalar Potential: $\phi = \mathbf{y^2 \sin x + xz^3 - 4y + 2z + C}$.

University Solved Examination Questions (Unit 3)

MMDU Official University Solved Examination Questions (Unit 3)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 7 (1 Mark): Homogeneous Degree via Euler's Theorem

Question:
Find the degree of homogeneity for $u(x,y) = \frac{x^3+y^3}{x-y}$ and evaluate $xu_x + yu_y$.

Step-by-Step Solution:
Since $u(tx, ty) = \frac{t^3(x^3+y^3)}{t(x-y)} = t^2 u(x,y)$, the degree is $n = 2$. By Euler's theorem, $xu_x + yu_y = 2u$.

🎯 **Verified Result:** Degree $n = 2$ | $xu_x + yu_y = 2u$.

📘 Question 19 (4 Marks): Two-Variable Extrema Determination

Question:

Find and classify all local extrema and saddle points of $f(x,y) = x^3 + y^3 - 3axy$ $(a > 0)$.

Step-by-Step Solution:

Stationary system: $f_x = 3x^2 - 3ay = 0$ and $f_y = 3y^2 - 3ax = 0 \implies$ critical points $(0,0)$ and (a,a).
With $r = 6x, s = -3a, t = 6y$ and $\Delta = rt - s^2 = 36xy - 9a^2$:

- At $(0,0)$: $\Delta = -9a^2 < 0 \implies$ **Saddle Point**.

- At (a,a): $\Delta = 27a^2 > 0, r = 6a > 0 \implies$ **Local Minimum** with $f_{\min} = -a^3$.

🎯 **Verified Result:** $(0,0)$ is a Saddle Point; (a,a) is a Local Minimum with value $-a^3$.

📘 Question 23 (6 Marks): Constrained Optimization via Lagrange Multipliers

Question:

Find the maximum and minimum distances from the origin to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ $(a > b > c > 0)$.

Step-by-Step Solution:

Using $F = (x^2 + y^2 + z^2) + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$, setting $F_x = F_y = F_z = 0$ yields $\lambda = -a^2, -b^2, -c^2$, corresponding to distances $d_{\max} = a$ and $d_{\min} = c$.

🎯 **Verified Result:** Maximum distance $= a$ | Minimum distance $= c$.